

# AfyaPlus Stakeholder Memo

AI Service Platform —Release v1.0.0

## What Was Built

Secure FastAPI Service → Docker Containerisation → MCP Server → LangChain Agent Integration

| Component          | Delivered capability                                                                                                                      |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Secure FastAPI API | JWT authentication, token expiry, role-based authorization, request validation and automated tests.                                       |
| Docker             | Containerised application, semantic versioning, Git/Docker release traceability and runtime secret injection.                             |
| MCP Server         | Logistics tools including stock checking, route planning and delivery ETA, with input validation, instructive error handling and logging. |
| LangChain Agent    | Multi-step tool use through MCP, authenticated API access and grounding controls for unsupported questions.                               |

## Highest-Value Component

The **LangChain agent + MCP integration** provides the greatest business value. It transforms the platform from a conventional API into an AI system that can orchestrate multiple operational tools to answer multi-step logistics questions. Grounding controls also reduce the risk of the agent confidently answering beyond the available data.

## Key Risk and Mitigation

**Risk:** The AI agent could generate a plausible but unsupported answer, potentially leading to an incorrect operational decision.

**Mitigation:** Grounding controls require unsupported questions to return “**The data cannot answer this.**” MCP tools validate inputs and return instructive errors. Automated testing and logging provide evidence that these controls operate as intended.

## Recommendation

**GO — for controlled deployment, with conditions.**

The platform is ready for the next controlled stage provided that: (1) secrets remain runtime-injected and are never committed or baked into images; (2) authentication, authorization, MCP validation and grounding tests remain part of releases; (3) Git and Docker semantic versions remain aligned; (4) end-to-end observability is strengthened so an API request can be correlated through the agent and MCP calls; and (5) the initial deployment is monitored and controlled rather than treated as unrestricted autonomous operation.